CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

AZEVEDO DAIRY #4

1257 W ROOSEVELT RD

EL NIDO CA 95317

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Antonio Azevedo Dairy # 4

Physical address of dairy:

Number and Street	City	County	Zip Code
1257 W Roosevelt RD	El Nido	Merced	95317

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/17/1988

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X074-X110-X026-XXXX

B. OPERATORS

Azevedo, Antonio or Alexandrina

Operator name: Azevedo, Antonio or Alexandrina

Telephone no.: (209) 723-0562 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2025 W El Nido Rd RD	El Nido	CA	95317

This operator is responsible for paying permit fees.

C. OWNERS

Azevedo, Antonio or Alexandrina

Legal owner name: Azevedo, Antonio or Alexandrina

Telephone no.: (209) 723-0562 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2025 W El Nido Rd RD	El Nido	CA	95317

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	800	300	0	0
Number under roof	430	1	0	0	0	0
Maximum number	1,100	1	800	300	0	0
Average number	765	1	800	300	0	0
Avg live weight (lbs)	1,200	1,250	875	550		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 66 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 30,002.82 *tons per reporting period*

Total nitrogen from manure: 356,442.83 *lbs per reporting period*

After ammonia losses (30% loss applied): 249,509.98 *lbs per reporting period*

Total phosphorus from manure: 60,261.77 *lbs per reporting period*

Total potassium from manure: 144,494.14 *lbs per reporting period*

Total salt from manure: 360,430.20 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,936,000 *gallons*

Total nitrogen generated: 33,773.00 *lbs*

Total phosphorus generated: 9,795.11 *lbs*

Total potassium generated: 40,400.08 *lbs*

Total salt generated: 252,472.97 *lbs*

	12,936,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,936,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/09/2023	Separator solids	1,488.00 *ton*	Dry-weight	67.8		18,600.00	2,300.00	3,300.00		0.00
08/10/2023	Corral solids	744.00 *ton*	Dry-weight	40.2		25,900.00	7,400.00	40,400.00		0.00
11/10/2023	Corral solids	3,528.00 *ton*	Dry-weight	37.9		25,600.00	7,400.00	41,800.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	153,043.77	41,213.87	222,269.42	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	153,043.77	41,213.87	222,269.42	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	35	35	3	process wastewater	X074-X110-X026-XXXX
Field 2/3	35	35	2	process wastewater	X074-X110-X026-XXXX
Heifer Ranch	80	80	2	manure	X074-X110-X033-XXXX
Totals for areas that were used for application	150	150	7		
Totals for areas that were not used for application					
Land application area totals	150	150	7		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	583.00 *ton*	Dry-weight		83.0	31,800.00	3,900.00	37,200.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.66	180.10	22.09	210.68	673.95

05/18/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 35 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/04/2023	210.00 *ton*	Dry-weight		18.3	11,200.00	3,600.00	40,800.00		13.10
10/23/2023	318.00 *ton*	Dry-weight		72.3	13,600.00	2,700.00	17,800.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	15.09	178.26	48.88	489.60	1,697.07

Field 2/3

Field name: Field 2/3

09/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 09/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	572.00 *ton*	Dry-weight		69.1	21,500.00	2,700.00	17,100.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.34	217.15	27.27	172.71	848.39

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	1,123.00 *ton*	Dry-weight		77.2	14,800.00	2,800.00	25,700.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.09	216.54	40.97	376.02	1,448.48

Heifer Ranch

Field name: Heifer Ranch

09/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	1,360.00 *ton*	Dry-weight		75.8	28,100.00	5,300.00	49,300.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.00	231.21	43.61	405.64	1,217.74

Heifer Ranch

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	2,368.00 *ton*	Dry-weight		69.0	13,600.00	2,700.00	12,800.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.60	249.59	49.55	234.91	1,376.40

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/22/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	69.49	29.99	93.66	691.14	1,584,000.00 *gal*
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Application event totals		70.10	29.99	93.66	691.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	77.21	18.43	76.10	524.96	792,000.00 *gal*
Irrigation Well	Ground water	0.44	0.00	0.00	0.00	4,620,000.00 *gal*
Application event totals		77.66	18.43	76.10	524.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	77.21	18.43	76.10	524.96	792,000.00 *gal*
Irrigation Well	Ground water	0.44	0.00	0.00	0.00	4,620,000.00 *gal*
Application event totals		77.66	18.43	76.10	524.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Application event totals		0.60	0.00	0.00	0.00	

Field 1 - 05/18/2023: Sudangrass, silage

Field 1 - 05/18/2023: Sudangrass, silage

Field name: Field 1

Crop: Sudangrass, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	89.81	22.08	88.12	584.13	1,056,000.00 *gal*
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	19.96	2,990,133.00 *gal*
Application event totals		90.41	22.08	88.12	604.09	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	49.22	15.08	72.54	377.67	660,000.00 *gal*
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	13.37	2,002,656.00 *gal*
Application event totals		49.83	15.08	72.54	391.04	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	49.22	15.08	72.54	377.67	660,000.00 *gal*
Irrigation Well	Ground water	0.45	0.00	0.00	0.00	4,752,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.80	3,265,518.00 *gal*
Application event totals		49.68	15.08	72.54	399.47	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	49.22	15.08	72.54	377.67	660,000.00 *gal*
Irrigation Well	Ground water	0.26	0.00	0.00	0.00	2,772,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	23.43	3,509,943.00 *gal*
Application event totals		49.49	15.08	72.54	401.10	

Field 1 - 05/18/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.45	0.00	0.00	0.00	4,752,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	24.98	3,741,332.00 *gal*
Application event totals		0.45	0.00	0.00	24.98	

Field 2/3 - 09/28/2022: Oats, silage-soft dough

Field name: Field 2/3

Crop: Oats, silage-soft dough Plant date: 09/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	69.49	29.99	93.66	691.14	1,584,000.00 *gal*
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Application event totals		70.10	29.99	93.66	691.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	77.21	18.43	76.10	524.96	792,000.00 *gal*
Irrigation Well	Ground water	0.44	0.00	0.00	0.00	4,620,000.00 *gal*
Application event totals		77.66	18.43	76.10	524.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	77.21	18.43	76.10	524.96	792,000.00 *gal*
Irrigation Well	Ground water	0.44	0.00	0.00	0.00	4,620,000.00 *gal*
Application event totals		77.66	18.43	76.10	524.96	

Field 2/3 - 06/18/2023: Corn, silage

Field 2/3 - 06/18/2023: Corn, silage

Field name: Field 2/3

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	112.26	27.60	110.15	730.16	1,320,000.00 *gal*
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	19.96	2,990,133.00 *gal*
Application event totals		112.87	27.60	110.15	750.13	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	88.60	27.14	130.58	679.81	1,188,000.00 *gal*
Irrigation Well	Ground water	0.60	0.00	0.00	0.00	6,336,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	13.37	2,002,656.00 *gal*
Application event totals		89.21	27.14	130.58	693.18	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	78.76	24.12	116.07	604.27	1,056,000.00 *gal*
Irrigation Well	Ground water	0.45	0.00	0.00	0.00	4,752,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	21.80	3,265,518.00 *gal*
Application event totals		79.21	24.12	116.07	626.07	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.26	0.00	0.00	0.00	2,772,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	23.43	3,509,943.00 *gal*
Application event totals		0.26	0.00	0.00	23.43	

Field 2/3 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.45	0.00	0.00	0.00	4,752,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	24.98	3,741,332.00 *gal*
Application event totals		0.45	0.00	0.00	24.98	

Heifer Ranch - 09/30/2022: Oats, silage-soft dough

Field name: Heifer Ranch

Crop: Oats, silage-soft dough

Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	291.70	71.69	217.54	0.00	1,920.00 *ton*
Application event totals		291.70	71.69	217.54	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.55	0.00	0.00	0.00	13,200,000.00 *gal*
Application event totals		0.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.50	0.00	0.00	0.00	11,880,000.00 *gal*
Application event totals		0.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.67	0.00	0.00	0.00	16,104,000.00 *gal*
Application event totals		0.67	0.00	0.00	0.00	

Heifer Ranch - 05/22/2023: Corn, silage

Field name: Heifer Ranch

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	54.28	18,584,448.00 *gal*
Application event totals		0.00	0.00	0.00	54.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	311.44	38.51	55.26	0.00	2,080.00 *ton*
Application event totals		311.44	38.51	55.26	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.18	0.00	0.00	0.00	4,356,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	29.42	10,073,243.00 *gal*
Application event totals		0.18	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.67	0.00	0.00	0.00	16,104,000.00 *gal*
Application event totals		0.67	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.55	0.00	0.00	0.00	13,200,000.00 *gal*
Application event totals		0.55	0.00	0.00	0.00	

Heifer Ranch - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	41.97	14,369,257.00 *gal*
Application event totals		0.00	0.00	0.00	41.97	

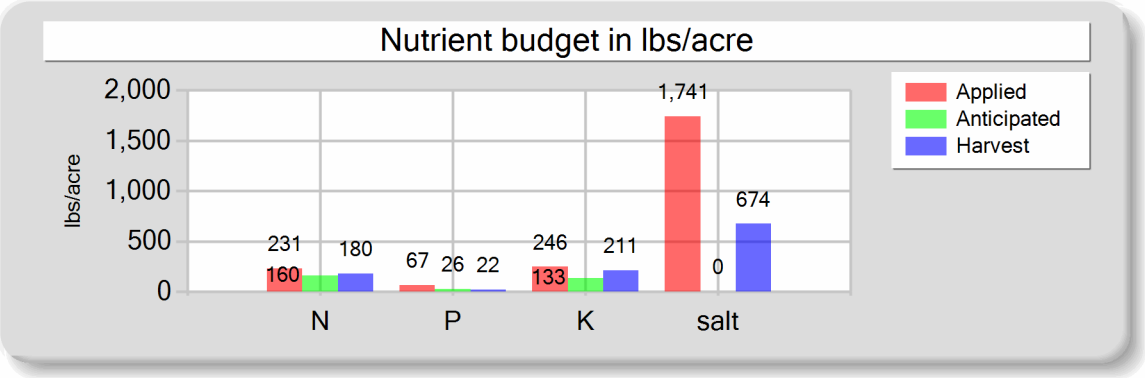
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.31	0.00	0.00	0.00	7,392,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	35.91	12,295,229.00 *gal*
Application event totals		0.31	0.00	0.00	35.91	

B. NUTRIENT BUDGET

Field 1 - 10/22/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.92	66.85	245.86	1,741.06
Fresh water	2.09	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	230.68	66.85	245.86	1,741.06
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	180.10	22.09	210.68	673.95
Nutrient balance	50.58	44.76	35.18	1,067.11
Applied to removed ratio	1.28	3.03	1.17	2.58

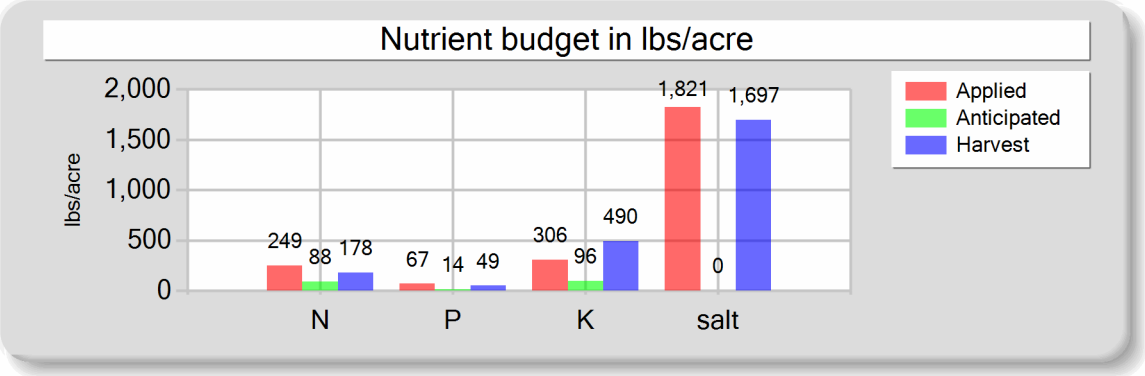
Fresh water applied	
21,912,000.00	*gallons*
806.94	*acre-inches*
23.06	*inches/acre*

Process wastewater applied	
3,168,000.00	*gallons*
116.67	*acre-inches*
3.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/18/2023: Sudangrass, silage

Field name: Field 1 Crop: Sudangrass, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	237.48	67.31	305.76	1,717.14
Fresh water	2.38	0.00	0.00	103.54
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	249.19	67.31	305.76	1,820.69
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	178.26	48.88	489.60	1,697.07
Nutrient balance	70.93	18.42	-183.84	123.62
Applied to removed ratio	1.40	1.38	0.62	1.07

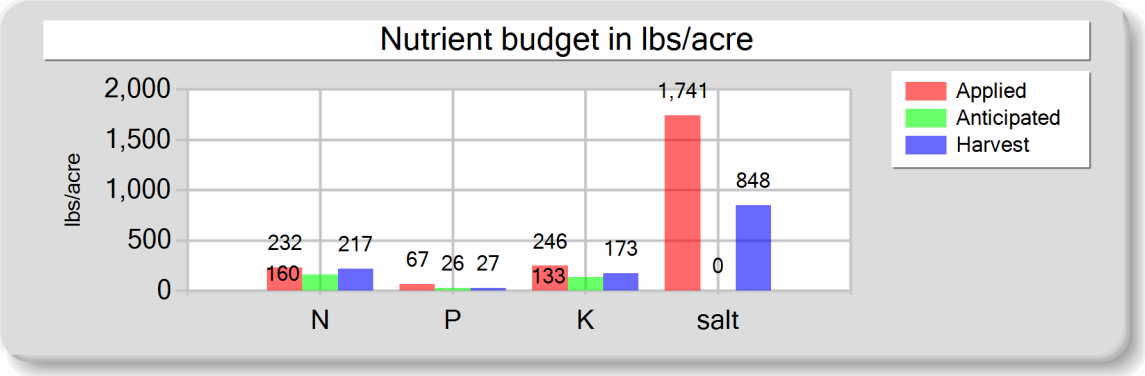
Fresh water applied	
40,457,582.00	*gallons*
1,489.92	*acre-inches*
42.57	*inches/acre*

Process wastewater applied	
3,036,000.00	*gallons*
111.81	*acre-inches*
3.19	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field 2/3 - 09/28/2022: Oats, silage-soft dough

Field name: Field 2/3 Crop: Oats, silage-soft dough Plant date: 09/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.92	66.85	245.86	1,741.06
Fresh water	1.49	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.41	66.85	245.86	1,741.06
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	217.15	27.27	172.71	848.39
Nutrient balance	15.26	39.58	73.16	892.67
Applied to removed ratio	1.07	2.45	1.42	2.05

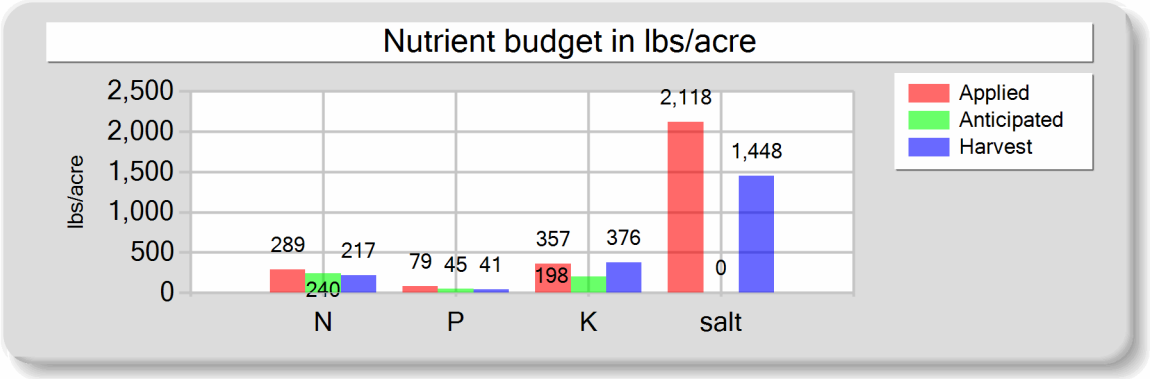
Fresh water applied	
15,576,000.00	*gallons*
573.61	*acre-inches*
16.39	*inches/acre*

Process wastewater applied	
3,168,000.00	*gallons*
116.67	*acre-inches*
3.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2/3 - 06/18/2023: Corn, silage

Field name: Field 2/3 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	279.62	78.86	356.80	2,014.24
Fresh water	2.38	0.00	0.00	103.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.00	78.86	356.80	2,117.79
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	216.54	40.97	376.02	1,448.48
Nutrient balance	72.46	37.89	-19.21	669.31
Applied to removed ratio	1.33	1.92	0.95	1.46

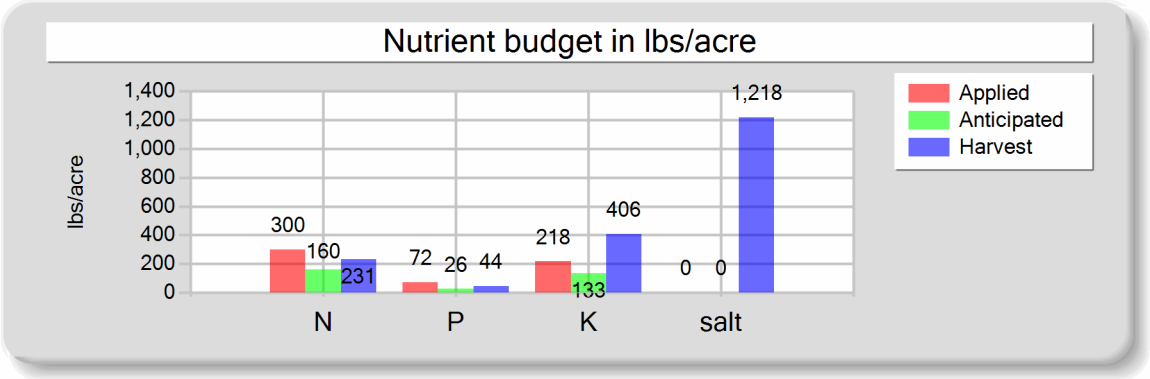
Fresh water applied	
40,457,582.00	*gallons*
1,489.92	*acre-inches*
42.57	*inches/acre*

Process wastewater applied	
3,564,000.00	*gallons*
131.25	*acre-inches*
3.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer Ranch - 09/30/2022: Oats, silage-soft dough

Field name: Heifer Ranch | Crop: Oats, silage-soft dough | Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	291.70	71.69	217.54	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.72	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.41	71.69	217.54	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	231.21	43.61	405.64	1,217.74
Nutrient balance	69.21	28.08	-188.10	-1,217.74
Applied to removed ratio	1.30	1.64	0.54	0.00

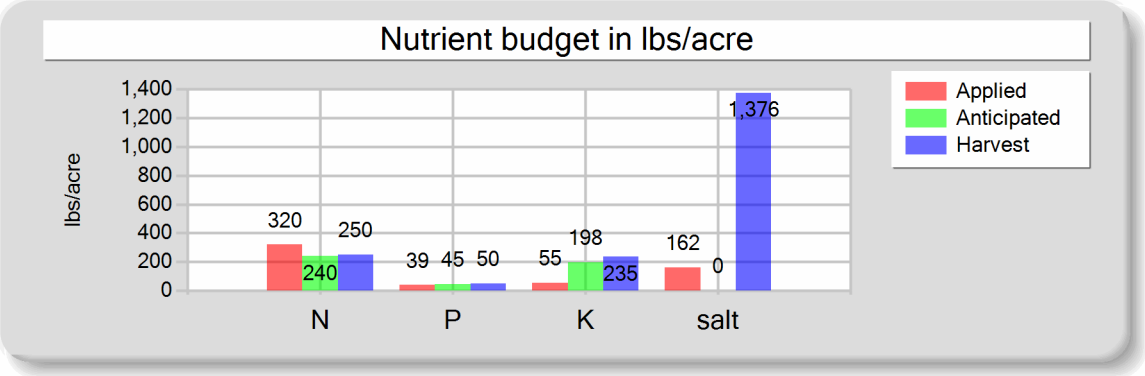
Fresh water applied	
41,184,000.00	*gallons*
1,516.67	*acre-inches*
18.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer Ranch - 05/22/2023: Corn, silage

Field name: Heifer Ranch Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	311.44	38.51	55.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.71	0.00	0.00	161.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.15	38.51	55.26	161.58
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.59	49.55	234.91	1,376.40
Nutrient balance	70.56	-11.04	-179.65	-1,214.82
Applied to removed ratio	1.28	0.78	0.24	0.12

Fresh water applied	
96,374,177.00	*gallons*
3,549.13	*acre-inches*
44.36	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022 Manure

Sample and source description: Fall 2022 Manure

Sample date: 10/10/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,900.00	8,800.00							
DL	1.00	1.00	1.00							

Separator manure

Sample and source description: Separator manure

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	2,300.00	3,300.00							
DL	1.00	1.00	1.00							

Dairy Corral Manure

Sample and source description: Dairy Corral Manure

Sample date: 10/06/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,600.00	7,400.00	41,800.00							0.00
DL	0.01	0.01	0.01							1.00

Heifers Corral Manure

Sample and source description: Heifers Corral Manure

Sample date: 10/06/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	7,400.00	40,400.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 12/06/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 6.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	182.00	172.00	0.00	2.00	79.40	248.00								4,000.00	1,830
DL	1.00	1.00	0.01	1.00	1.00	1.00								1.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	408.00	257.00	0.00	0.90	97.60	403.00								4,810.00	2,780
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	355.00	250.00	0.00	1.70	87.70	350.00								4,750.00	2,320
DL	1.00	1.00	0.01	1.00	1.00	1.00								1.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	312.00	257.00	0.00	0.80	95.80	461.00								4,890.00	2,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	339.00	191.00	0.00	0.00	106.00	556.00								5,000.00	3,050
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

Canal

MID

Sample description: MID

Sample date: 10/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								32.20	28
DL	0.01	0.01	0.01								1.00	1

Domestic Well

DW Heifers

Sample description: DW Heifers

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.20	0.00	11.20								1,400.00	0
DL	0.01	0.01	0.01								10.00	1

DW House

Sample description: DW House

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.40	0.00	1.40								215.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.00	0.00	2.00								278.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW

Sample description: IW

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.40	0.00	0.40								94.20	0
DL	0.01	0.01	0.01								10.00	1

IW Heifers

Sample description: IW Heifers

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.40	0.00	0.40								73.30	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/22/2022: Oats, silage-soft dough

Field 1 Oat Silage

Sample and source description: Field 1 Oat Silage

Sample date: 04/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,800.00	3,900.00	37,200.00		11.90
DL	0.01	0.01	0.01		1.00

Field 1 - 05/18/2023: Sudangrass, silage

Field 1 Sudan Hay

Sample and source description: Field 1 Sudan Hay

Sample date: 08/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	3,600.00	40,800.00		13.10
DL	0.01	0.01	0.01		1.00

Field 1 Sudan 2nd Cut

Sample and source description: Field 1 Sudan 2nd Cut

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,700.00	17,800.00		8.20
DL	0.01	0.01	0.01		1.00

Field 2/3 - 09/28/2022: Oats, silage-soft dough

Field 2/3 - 09/28/2022: Oats, silage-soft dough

Oats Field 2/3

Sample and source description: Oats Field 2/3

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	2,700.00	17,100.00		8.40
DL	0.01	0.01	0.01		1.00

Field 2/3 - 06/18/2023: Corn, silage

Field 2/3 Corn

Sample and source description: Field 2/3 Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	2,800.00	25,700.00		9.90
DL	0.01	0.01	0.01		1.00

Heifer Ranch - 09/30/2022: Oats, silage-soft dough

Heifers Oat Silage

Sample and source description: Heifers Oat Silage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,100.00	5,300.00	49,300.00		14.80
DL	0.01	0.01	0.01		1.00

Heifer Ranch - 05/22/2023: Corn, silage

Field 46 Corn

Sample and source description: Field 46 Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,700.00	12,800.00		7.50
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

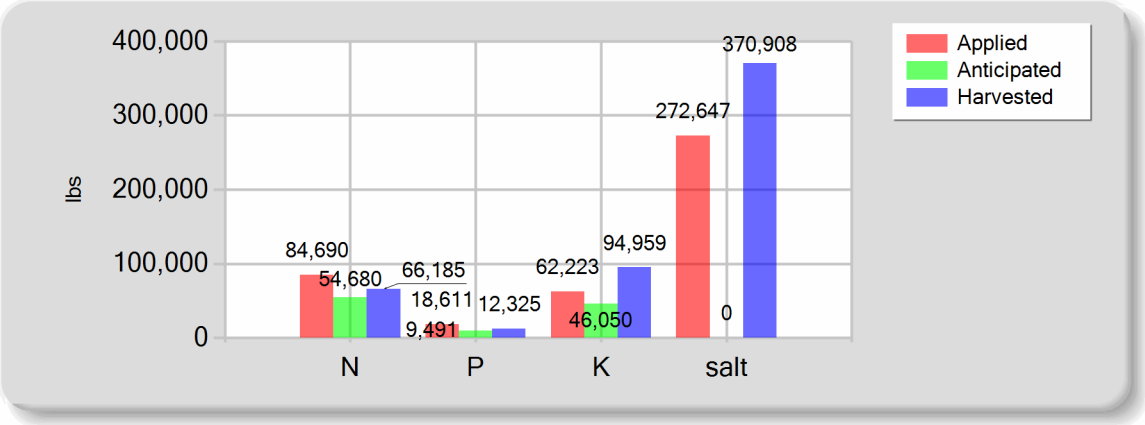
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

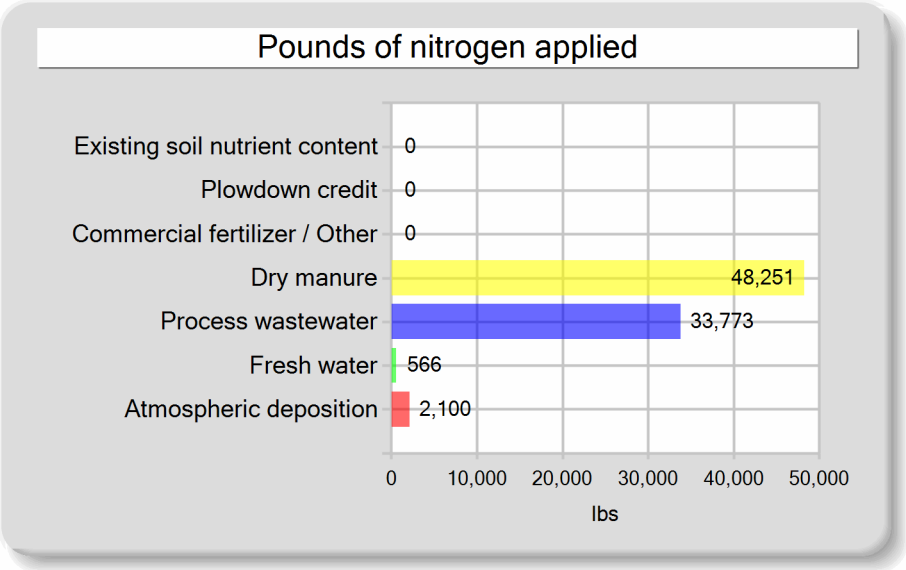
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

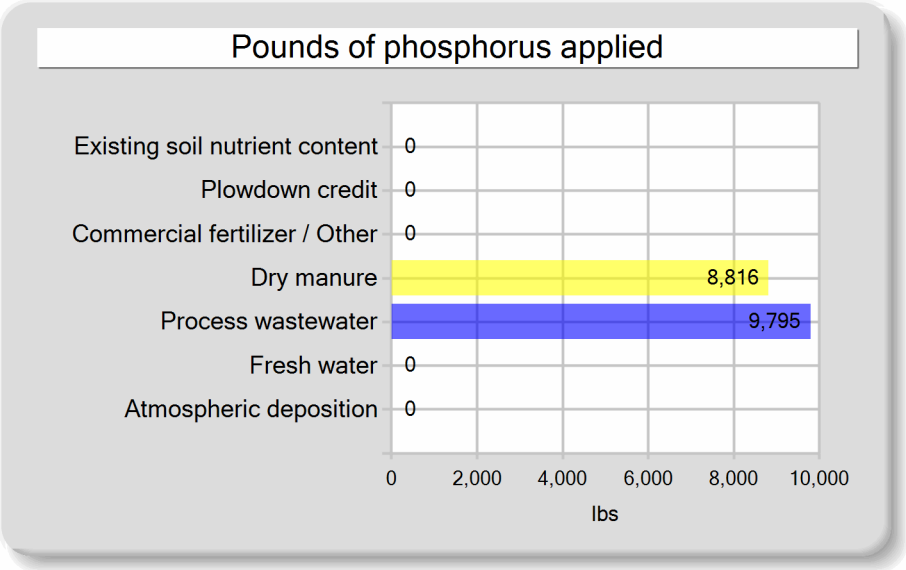
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	48,250.75	8,815.94	21,823.30	0.00
Process wastewater	33,773.00	9,795.11	40,400.08	252,472.97
Fresh water	566.19	0.00	0.00	20,174.52
Atmospheric deposition	2,100.00	0.00	0.00	0.00
Total nutrients applied	84,689.94	18,611.05	62,223.38	272,647.49
Anticipated crop nutrient removal	54,680.00	9,491.00	46,050.00	0.00
Actual crop nutrient removal	66,185.09	12,325.02	94,958.88	370,907.52
Nutrient balance	18,504.85	6,286.03	-32,735.50	-98,260.03
Applied to removed ratio	1.28	1.51	0.66	0.74

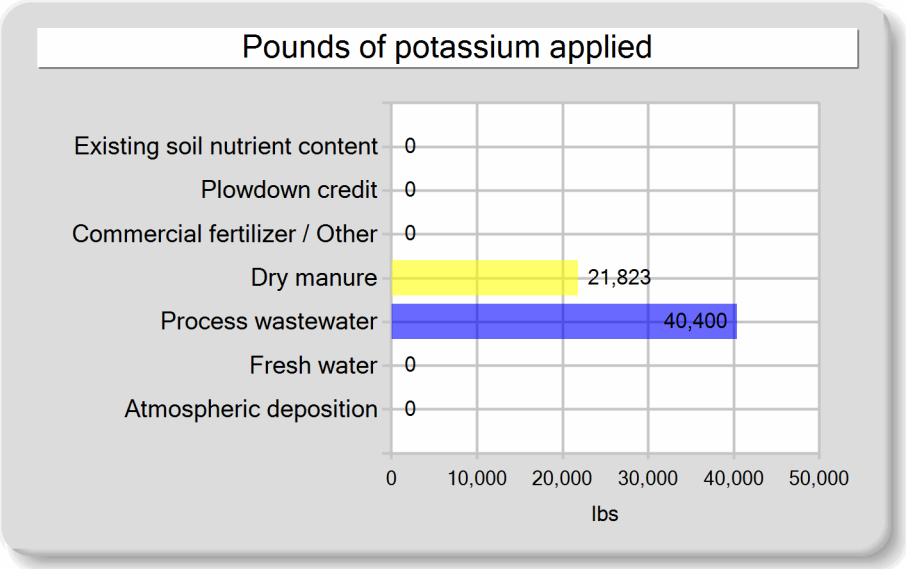
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

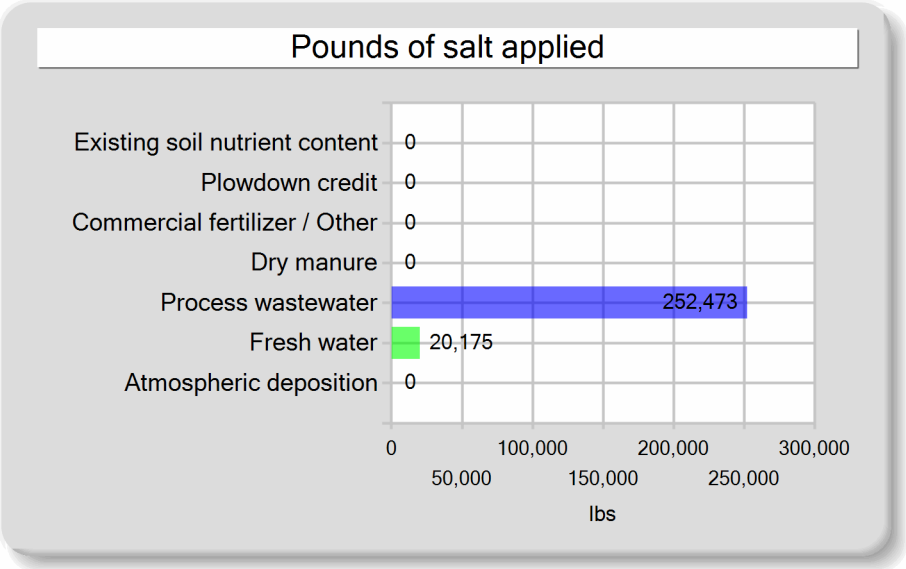


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Antonio Azevedo

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Antonio or Alexandrina Azevedo	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Antonio Azevedo

Name of Dairy Facility: Antonio Azevedo Dairy # 4

Facility Address:

Number and Street	City	County	Zip Code
1257 W Roosevelt RD	El Nido	Merced	95317

Contact Person Name and Phone Number:

Name	Phone Number
Antonio Azevedo	(209) 230-8765

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Gomes Farms LLC

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 149	El Nido	CA	95317

Contact Person:

Name	Phone Number
Eddie Gomes	(209) 564-2431

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Gomes Farms LLC	(209) 564-2431

Address	City	State	Zip Code
P.O. Box 149	El Nido	CA	95317

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	El Nido	95317

Street and nearest cross street (if no address)	County
Coyote Rd	Merced

Assessor's Parcel Number	Assessor's Parcel Number County
X074-X160-X054-XXXX	Merced

Last date hauled: 08/09/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,488.00 tons

Manure Solids Content: 32.2 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Antonio Azevedo 8/9/23

Operator Signature Date

ESE ATTACHED

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,488.00 tons

Manure Solids Content: 32.2 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

SEE ATTACHED 8/9/23

Operator Signature Date

Eddie Gomes 8/9/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Antonio Azevedo

Name of Dairy Facility: Antonio Azevedo Dairy # 4

Facility Address:

Number and Street	City	County	Zip Code
1257 W Roosevelt RD	El Nido	Merced	95317

Contact Person Name and Phone Number:

Name	Phone Number
Antonio Azevedo	(209) 230-8765

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Gomes Farms LLC

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 149	El Nido	CA	95317

Contact Person:

Name	Phone Number
Eddie Gomes	(209) 564-2431

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Gomes Farms LLC	(209) 564-2431

Address	City	State	Zip Code
P.O. Box 149	El Nido	CA	95317

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	El Nido	95317

Street and nearest cross street (if no address)	County
Coyote Rd	Mendocino

Assessor's Parcel Number	Assessor's Parcel Number County
X074-X160-X054-XXXX	Merced

Last date hauled: 08/10/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 744.00 tons

Manure Solids Content: 59.8 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Antonio Azevedo 8/10/23

Operator Signature Date

SEE ATTACHED

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 744.00 tons

Manure Solids Content: 59.8 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

SEE ATTACHED — 8/10/23

Operator Signature — Date

Eddie Gomes — 8/10/23

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Antonio Azevedo

Name of Dairy Facility: Antonio Azevedo Dairy # 4

Facility Address:

Number and Street	City	County	Zip Code
1257 W Roosevelt RD	El Nido	Merced	95317

Contact Person Name and Phone Number:

Name	Phone Number
Antonio Azevedo	(209) 230-8765

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Gomes Farms LLC

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 149	El Nido	CA	95317

Contact Person:

Name	Phone Number
Eddie Gomes	(209) 564-2431

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Gomes Farms LLC	(209) 564-2431

Address	City	State	Zip Code
P.O. Box 149	El Nido	CA	95317

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	El Nido	95317

Street and nearest cross street (if no address)	County
Coyote Rd	Merced

Assessor's Parcel Number	Assessor's Parcel Number County
X074-X160-X054-XXXX	Merced

Last date hauled: 11/10/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 3,528.00 tons

Manure Solids Content: 62.1 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Antonio Azevedo 11/10/23

Operator Signature Date

SEE ATTACHED

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 3,528.00 tons

Manure Solids Content: 62.1 %

Method used to determine amount of manure:

tons per load per hauler

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

SEE ATTACHED

11/10/23

Operator Signature — Date

Eddie Gomes

11/10/23

Hauler Signature — Date

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido CA 95317
00-0021345 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0781
Sampled By: N Machado
Requested By: Antonio & Alexandria Azevedo
Submitted Date: 12/13/23
Reported Date: 12/20/23
Project: Azevedo Dairy #4
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/11/23	20:00	278	0.28	2.0	ND		7.3
2	DW - House	12/11/23	20:15	215	0.22	1.4	ND		7.4
3	IW	12/11/23	19:15	94.2	0.09	0.4	ND		7.3
4	DW - Heifers	12/11/23	19:45	1400	1.40	11.2	ND		7.5
5	IW - Heifers	12/11/23	19:30	73.3	0.07	0.4	ND		7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido CA 95317
00-0021345 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0781
Sampled By: N Machado
Requested By: Antonio & Alexandria Azevedo
Submitted Date: 12/13/23
Reported Date: 12/20/23
Project: Azevedo Dairy #4
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/11/23	20:00	278	0.28	**2.0**	5.46	0.46
2	DW - House	12/11/23	20:15	215	0.22	1.4	3.82	0.32
3	IW	12/11/23	19:15	94.2	0.09	0.4	1.09	0.09
4	DW - Heifers	12/11/23	19:45	**1400**	**1.40**	**11.2**	30.58	2.55
5	IW - Heifers	12/11/23	19:30	73.3	0.07	0.4	1.09	0.09

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0781-01	DW - MB	Ag Water	N Machado		12/11/2023 20:00
23L0781-02	DW - House	Ag Water	N Machado		12/11/2023 20:15
23L0781-03	IW	Ag Water	N Machado		12/11/2023 19:15
23L0781-04	DW - Heifers	Ag Water	N Machado		12/11/2023 19:45
23L0781-05	IW - Heifers	Ag Water	N Machado		12/11/2023 19:30

Default Cooler Temperature on Receipt °C: 2.0
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Sample Results

Sample: DW - MB
23L0781-01 (Water)

Sampled: 12/11/2023 20:00
Sampled By: N Machado

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.28**	mmhos/cm	0.01	1		12/14/23 17:44	SM 2510 B		BEL0645
Electrical Conductivity umhos	**278**	umhos/cm	10.0	1		12/14/23 17:44	SM 2510 B		BEL0645
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 20:00	Field		BEL0450
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	12/14/23 02:57	EPA 300.0		BEL0447
Temperature	**25.0**	units	0.0	1		12/14/23 17:44	SM 4500-H+	H	BEL0645
pH	**7.3**	units	1.0	1		12/14/23 17:44	SM 4500-H+	H	BEL0645

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Sample Results
(Continued)

Sample: DW - House
23L0781-02 (Water)

Sampled: 12/11/2023 20:15
Sampled By: N Machado

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.22**	mmhos/cm	0.01	1		12/14/23 17:45	SM 2510 B		BEL0645
Electrical Conductivity umhos	**215**	umhos/cm	10.0	1		12/14/23 17:45	SM 2510 B		BEL0645
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 20:15	Field		BEL0450
Nitrate Nitrogen as NO3N	**1.4**	mg/L	0.1	1	10	12/14/23 03:18	EPA 300.0		BEL0447
Temperature	**25.0**	units	0.0	1		12/14/23 17:45	SM 4500-H+	H	BEL0645
pH	**7.4**	units	1.0	1		12/14/23 17:45	SM 4500-H+	H	BEL0645

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Sample Results
(Continued)

Sample: IW
23L0781-03 (Water)

Sampled: 12/11/2023 19:15
Sampled By: N Machado

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.09**	mmhos/cm	0.01	1		12/14/23 17:46	SM 2510 B		BEL0645
Electrical Conductivity umhos	**94.2**	umhos/cm	10.0	1		12/14/23 17:46	SM 2510 B		BEL0645
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 19:15	Field		BEL0450
Nitrate Nitrogen as NO3N	**0.4**	mg/L	0.1	1	10	12/14/23 03:39	EPA 300.0		BEL0447
Temperature	**25.0**	units	0.0	1		12/14/23 17:46	SM 4500-H+	H	BEL0645
pH	**7.3**	units	1.0	1		12/14/23 17:46	SM 4500-H+	H	BEL0645

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Sample Results
(Continued)

Sample: DW - Heifers
23L0781-04 (Water)

Sampled: 12/11/2023 19:45
Sampled By: N Machado

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.40**	mmhos/cm	0.01	1		12/14/23 17:48	SM 2510 B		BEL0645
Electrical Conductivity umhos	**1400**	umhos/cm	10.0	1		12/14/23 17:48	SM 2510 B		BEL0645
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 19:45	Field		BEL0450
Nitrate Nitrogen as NO3N	**11.2**	mg/L	0.1	1	10	12/14/23 04:00	EPA 300.0		BEL0447
Temperature	**25.0**	units	0.0	1		12/14/23 17:48	SM 4500-H+	H	BEL0645
pH	**7.5**	units	1.0	1		12/14/23 17:48	SM 4500-H+	H	BEL0645

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Sample Results
(Continued)

Sample: IW - Heifers
23L0781-05 (Water)

Sampled: 12/11/2023 19:30
Sampled By: N Machado

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.07**	mmhos/cm	0.01	1		12/14/23 17:49	SM 2510 B		BEL0645
Electrical Conductivity umhos	**73.3**	umhos/cm	10.0	1		12/14/23 17:49	SM 2510 B		BEL0645
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 19:30	Field		BEL0450
Nitrate Nitrogen as NO3N	**0.4**	mg/L	0.1	1	10	12/14/23 04:21	EPA 300.0		BEL0447
Temperature	**25.0**	units	0.0	1		12/14/23 17:49	SM 4500-H+	H	BEL0645
pH	**7.9**	units	1.0	1		12/14/23 17:49	SM 4500-H+	H	BEL0645

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0447										
Blank (BEL0447-BLK1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0447-BLK2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0447-BLK3)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0447-BLK4)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0447-BS1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.1	90-110		
LCS (BEL0447-BS2)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.5	90-110		
LCS (BEL0447-BS3)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.07		0.1	mg/L	5.000		1.44	90-110		
Duplicate (BEL0447-DUP1)		**Source: 23L0636-01**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			5.26	10
Duplicate (BEL0447-DUP2)		**Source: 23L0777-05**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	1.8		0.1	mg/L		1.8			0.112	10
Duplicate (BEL0447-DUP3)		**Source: 23L0681-01**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			2.74	10
Matrix Spike (BEL0447-MS1)		**Source: 23L0636-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000	0.04	96.0	90-110		
Matrix Spike (BEL0447-MS2)		**Source: 23L0777-05**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	6.7		0.1	mg/L	5.000	1.8	98.2	90-110		
Matrix Spike (BEL0447-MS3)		**Source: 23L0681-01**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	0.04	98.0	90-110		
Reference (BEL0447-SRM1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		99.7	90-110		
Reference (BEL0447-SRM2)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		98.5	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0447 (Continued)										
Reference (BEL0447-SRM3)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		98.2	90-110		
Reference (BEL0447-SRM4)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		95.5	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0645										
Blank (BEL0645-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0645-BLK2)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.8		1.0	units						
Blank (BEL0645-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Duplicate (BEL0645-DUP1)		**Source: 23L0784-02**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	2.01		0.01	mmhos/cm		2.01			0.105	10
pH	7.8		1.0	units		7.8			0.385	10
Electrical Conductivity umhos	2010		10.0	umhos/cm		2010			0.105	10
Duplicate (BEL0645-DUP2)		**Source: 23L0802-04**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.44		0.01	mmhos/cm		0.45			1.32	10
Electrical Conductivity umhos	442		10.0	umhos/cm		448			1.32	10
pH	9.1		1.0	units		9.1			0.329	10
Reference (BEL0645-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	422			umhos/cm	426.0		99.1	90-110		
Reference (BEL0645-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0645-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0645-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0645-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Antonio Azevedo Dairies
2025 W El Nido Road
El Nido, CA 95317

Account# 00-0021345
Account Manager: Ben Nydam
Submitted By: Antonio & Alexandria Azevedo
Ranch: Azevedo Dairy #4

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:33

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0645 (Continued)										
Reference (BEL0645-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0645-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.1			units	4.000		102	97.5-102.5		
Reference (BEL0645-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0645-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

12/13/23 07:00 23L0781

VG

WATER WORK REQUEST

Acct No. / Cons.

Bill To: 21345 | 8

Purchase Order No. | Results Needed By

Client: **Azevedo Dairies, Antonio**
Address: 2025 W El Nido Road
City, State, Zip: El Nido, CA 95317
Phone | Fax
Cell/Email

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Antonio & Alexandria Azevedo - (209) 261-4771

Ranch: Azevedo Dairy #4

Date sampled: 12/11/23

Sampled by: N Machado

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8175

No. of Samples: 5 No. Bottles: 5

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES	Sampled From:	Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C H/E
1. DW-MB		12/11/23	8pm	Ø	2.0 / 1.9
2. DW-House		12/11/23	815p	Ø	2.3 / 1.8
3. IW		12/11/23	715p	Ø	3.4 / .5
4. DW-Heifers		12/11/23	745p	Ø	3.8 / .4
5. IW-Heifers		12/11/23	730p	Ø	3.9 / .4
6.					
7.					
8.					
9.					
10.					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

IR Thermometer SN: 221511276
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Hanford

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	N Machado	Azevedo Dairies	12/11/23 815p	12/12/23 730A
Second	M Pedroso	CASIRE	12/12/23 730A	12/12/23 330P
Third	[illegible]	DLI	12/12/23 3:36pm	
Fourth	[illegible]	DLI	12/13 0800	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$ In
$ Out

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid | Rec By | Check No. | Date

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☐ Walk In ☑ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1								
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	+	+	1	1	1					
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/13/23 07:00 23L0781

SENDOUT